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Cath Lab Orientation Binder

Cath Lab RN-Focused Edition
Department: Cath Lab
Documentation Platform: Epic Cupid
Version: Updated Draft v2

Title and Ownership Page

Hospital / System: _____
Department: _____
Unit Educator: _____
Manager / Director: _____
Primary Epic Superuser / Builder: _____
Version Date: _____
Next Scheduled Review: _____

Purpose

This binder standardizes onboarding for Cath Lab nurses with supporting cardiologist-facing workflow considerations. It provides a structured reference for workflow, patient safety, medications, angiography, PCI support, complication response, Epic Cupid charting, and competency validation.

Intended Users

- New Cath Lab RNs
- New radiology technologists
- Cross-training procedural staff
- Preceptors
- Charge nurses and educators

Design Approach

This binder is written as a shared core orientation manual for both nurses and radiology techs. Sections are organized around actual case flow so both disciplines learn the same mental model of the procedure. Role-specific expectations are identified within each section and then reinforced again in the dedicated skills and competency tabs.

How to Use This Binder

This binder should be used in four ways: 1. As a week-by-week orientation guide 2. As a case-day quick reference 3. As a procedure documentation support tool 4. As a competency signoff record

Recommended Use by Audience

Orientee: review the section that matches the next expected case phase before the shift and use the quick references during live cases.
Preceptor: assign a daily focus, observe real performance, and document progress in the competency tools.
Educator or supervisor: use the signoff and weekly review tools to identify strengths, gaps, and readiness for independent workflow.

Binder Organization Logic

The binder moves in the same sequence as a real case: - orientation expectations - room flow and communication - pre-procedure preparation - intra-procedure monitoring - angiography and PCI workflow - complex cases and complications - Epic Cupid charting - role-specific competency validation

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1. Orientation Roadmap

Purpose

Provide a structured path from first-day observation to safe independent participation in routine Cath Lab workflow. This section sets expectations for the orientee, the preceptor, and the department so orientation progresses in a deliberate and measurable way.

Orientation Philosophy

Orientation should move from recognition, to guided performance, to independent routine workflow, and finally to complex-case readiness. The orientee is not expected to master every lesion subset or complication in the first phase. The expectation is progressive development of safe habits, reliable communication, sound charting, and anticipation of procedural flow.

Global Orientation Outcomes

At the completion of orientation, the orientee should be able to: - prepare the room and patient safely before the case begins - identify the major phases of diagnostic and interventional procedures - maintain active awareness of patient status, hemodynamics, ECG, radiation, contrast, access, and equipment position - perform role-specific charting in Epic Cupid in real time - recognize red flags early and escalate them immediately - complete routine Cath Lab workflow with minimal prompting - participate appropriately in higher-risk or more complex cases

Orientation Phases

Phase 1: Foundational Observation

Primary goal: understand room flow and major case phases.

Focus areas: - room setup and room turnover - sterile field awareness - patient transport and room entry - monitor hookup and baseline readiness - introduction to pressure waveforms and ECG surveillance - role identification for RN, radiology tech, physician, and documenter - introduction to Procedure Log navigation in Epic Cupid

Expected orientee behaviors: - identifies core room equipment and supply locations - follows the sequence of a routine case without losing track of the phase of care - knows who to notify

for patient status changes, equipment concerns, or documentation uncertainty - begins using appropriate procedural terminology

Preceptor focus: - explain case flow out loud - identify the reason behind each major step - introduce safety habits early - model concise call-outs and escalation language

Phase 2: Guided Participation

Primary goal: perform defined tasks with direct preceptor support.

Focus areas: - pre-procedure preparation - baseline patient assessment and readiness checks - medication setup and medication awareness - real-time monitoring responsibilities - access-site awareness - charting in parallel with the preceptor - routine diagnostic angiography support - simple radial and femoral cases

Expected orientee behaviors: - assists with room setup and patient preparation - completes selected documentation elements with supervision - recognizes abnormal ECG or pressure changes and verbalizes concern - begins anticipating the next step in routine cases

Preceptor focus: - assign limited but real tasks - correct workflow habits immediately - reinforce charting sequence in Epic Cupid - emphasize timing of escalation, not just recognition of problems

Phase 3: Routine Case Performance

Primary goal: manage routine workflow with minimal prompting.

Focus areas: - routine diagnostic cath workflow - standard PCI workflow - ACT timing and result awareness - radiation and contrast awareness - access-site surveillance and closure readiness - case-end documentation and handoff - independent charting of routine cases with review

Expected orientee behaviors: - manages assigned role with minimal prompting - charts routine workflow accurately and in real time - escalates patient, access, or equipment concerns appropriately - participates in room turnover and preparation for the next case

Preceptor focus: - allow the orientee to run a defined portion of routine workflow - audit for omissions rather than narrating every step - reinforce prioritization under time pressure - verify that the orientee can distinguish routine variation from true abnormality

Phase 4: Complex Case Readiness

Primary goal: safely support high-risk and higher-complexity procedures.

Focus areas: - bifurcation disease - left main intervention - calcified lesions - thrombotic ACS and STEMI - graft cases - TAVR reaccess cases - SCAD awareness - long lesions and diffuse disease - small or large vessel considerations - complication recognition and bailout readiness

Expected orientee behaviors: - anticipates higher-risk equipment and room setup needs - understands when a case is moving out of routine territory - identifies early signs of

complication and escalates promptly - supports complex cases without losing the overall procedural sequence

Preceptor focus: - teach anticipation rather than memorization - reinforce backup planning - explain why certain cases favor different access, imaging, or support strategies - emphasize complication prevention, not just complication treatment

Suggested Timeline

This may vary based on prior experience, staffing, and case volume.

- **Week 1–2:** observation and basic participation
- **Week 3–4:** routine case task performance and baseline charting
- **Week 5–8:** documentation independence and standard PCI support
- **Week 8+:** complex case support and final validation

Orientation Structure by Learning Domain

Throughout orientation, the orientee should progress in five domains at the same time: - patient and room safety - workflow recognition - monitoring and escalation - Epic Cupid documentation - role-specific technical performance

Weekly Orientation Priorities

Week 1

- room layout
- staff roles
- sterile awareness
- basic patient flow
- Epic Cupid screen familiarity

Week 2

- pre-procedure checklist participation
- baseline assessments
- basic monitoring
- procedure-log event recognition

Week 3

- charting routine case phases with guidance
- access and sheath awareness
- medication workflow observation and partial participation

Week 4

- independent documentation of defined sections
- case-start and case-end workflow accuracy
- postprocedure and handoff awareness

Week 5–6

- routine diagnostic case support
- simple intervention support
- contrast, ACT, and radiation tracking awareness

Week 7–8

- routine case independence
- complication escalation drills
- closure workflow
- competency audit and remediation if needed

Orientation Milestones

The orientee should demonstrate the following before advancing:

Milestone 1: Room and Safety Readiness

- locates core supplies
- understands room roles
- demonstrates sterile awareness
- recognizes pre-case safety steps

Milestone 2: Monitoring and Communication

- identifies abnormal vitals or ECG changes
- recognizes dampened waveform or access bleeding as urgent findings
- calls out critical changes without delay

Milestone 3: Charting Accuracy

- charts core case phases correctly in Epic Cupid
- distinguishes pre-procedure, intra-procedure, and postprocedure documentation
- completes case-end essentials before patient transfer

Milestone 4: Routine Workflow Independence

- performs role-specific tasks reliably in standard cases
- prioritizes workflow appropriately
- requires only limited prompting

Preceptor Expectations

Preceptors should: - define daily goals at the beginning of the shift - review one to three key learning points after each case - correct unsafe habits immediately - avoid overloading the orientee with too many simultaneous tasks early in training - document progress honestly, including areas needing remediation

Orientee Expectations

Orientees should: - arrive prepared and engaged - ask questions before guessing when safety is affected - chart in real time whenever possible - communicate concerns early - accept correction and incorporate feedback into the next case

Progress Review Template

Use at least weekly.

Strengths this week: _____
Safety concerns noted: _____
Charting gaps noted: _____
Procedural workflow gaps noted: _____
Focus for next week: _____
Preceptor initials/date: _____

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2. Room Roles, Workflow, and Communication

Purpose

Define team expectations so new staff understand room flow, role boundaries, communication standards, and escalation expectations.

Core Principle

The Cath Lab functions safely only when everyone knows the current case phase, the next expected step, and who owns each responsibility in that moment. Staff should communicate early, clearly, and specifically.

Standard Room Roles

Registered Nurse

Primary nursing responsibilities include: - pre-procedure patient assessment and readiness review - medication administration and medication safety monitoring - sedation assessment, monitoring, and reassessment - hemodynamic and symptom surveillance - ACT and anticoagulation awareness - access-site, sheath, and closure monitoring from the nursing perspective - nursing documentation in Epic Cupid - postprocedure assessment and handoff

Radiology Technologist

Primary radiology tech responsibilities include: - room and imaging system readiness - table setup and procedural supply organization - wire, catheter, and device readiness - radiation reduction support and image optimization - procedural anticipation and technical support -

sterile field protection - procedural documentation support as assigned locally - equipment turnover and room reset

Physician / Operator

Primary operator responsibilities include: - procedure planning and execution - access strategy - angiography interpretation - lesion crossing and treatment - management of complications - final procedural decisions and postprocedure plan

Documenter / Circulator Role

When this role is assigned separately, responsibilities may include: - procedure-log event support - timing awareness - staff role updates - procedure phase tracking - handoff support - ensuring documentation completion prior to transfer

Standard Workflow for a Routine Case

1. Before patient enters the room

- room setup complete
- equipment checked
- supplies available
- emergency equipment immediately accessible
- monitor and pressure systems ready
- Epic Cupid case opened and correct patient verified

2. Patient enters room

- patient identity verified
- staff roles confirmed
- patient connected to monitors
- baseline vitals and rhythm reviewed
- oxygen and pulse oximetry confirmed as needed
- site prep and safety steps completed

3. Before procedure start

- consent verified
- pre-procedure checklist complete
- timeout performed
- case start documented
- sedation plan confirmed if applicable

4. During procedure

- continuous monitoring maintained
- medications given and charted
- access and procedural events documented
- contrast and radiation awareness maintained
- significant changes called out immediately

5. End of case

- site condition assessed
- wires and catheters accounted for and documented
- contrast and radiation documented
- case end and sedation end completed if applicable
- handoff prepared and patient transitioned safely

Communication Standards

Communication in the Cath Lab should be brief, clear, and specific.

Appropriate call-out examples

- “New ST elevation.”
- “Pressure dampening.”
- “SpO2 dropping.”
- “ACT due.”
- “Contrast rising.”
- “Radiation threshold approaching.”
- “Site bleeding.”
- “Resistance advancing wire.”
- “Patient reports chest pain.”
- “Hypotension.”

Poor communication examples to avoid

- “Something looks off.”
- “I’m not sure but maybe…”
- delayed reporting after repeated observation of an abnormal change
- assuming someone else saw the problem

Escalation Triggers

Immediate verbal escalation is required for: - new chest pain or ischemic symptoms - new ST change or rhythm instability - sudden hypotension or hypertension of concern - dampened arterial waveform - oxygen desaturation or respiratory change - access-site bleeding, enlarging hematoma, or loss of pulse concern - severe patient agitation or oversedation - suspected dissection, perforation, or no-reflow - unexpected difficulty advancing or retrieving equipment - sterile break - equipment malfunction affecting safety

Pre-Case Huddle Template

Before every case, review: - patient identity - indication for procedure - planned procedure(s) - access strategy - anticoagulation and antiplatelet considerations - allergies and contrast concerns - renal function and contrast strategy - anticipated devices - anticipated complications or lesion complexity - closure plan - postprocedure destination

Role Clarity Questions for New Staff

At the start of each case, the orientee should be able to answer: - Who is documenting? - Who is giving medications? - Who is monitoring sedation? - Who is watching the access site? - Who is entering procedural devices or supplies? - Who is responsible for final case-end completion?

Common Workflow Failures

These should be reviewed during orientation: - poor task ownership - duplicate charting by multiple staff members - no one tracking radiation or contrast - late recognition of patient instability - failure to announce waveform dampening or ischemic changes - rushing transfer before chart completion - incomplete handoff

Handoff Expectations

The receiving team should be told at minimum: - procedure performed - access site and closure status - sedation received - immediate complications or concerns - postprocedure restrictions - required monitoring priorities - pending orders or follow-up tasks

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3. Pre-Procedure Preparation

Purpose

Standardize what must be completed before vascular access and before the first contrast injection so the case begins with the patient, staff, equipment, and documentation fully ready.

Core Principle

The safest case start is a deliberate one. Most avoidable problems in the room begin before the procedure starts, including wrong assumptions about access, missing baseline assessments, incomplete consent verification, missing equipment, or documentation gaps.

Pre-Procedure Goals

Before the case begins, the team should confirm: - the correct patient is in the correct room for the correct procedure - the patient is clinically ready - required equipment is available - staff understand the plan - documentation is current and accurate - anticipated complications have been considered

Patient Readiness Checklist

Identity and procedure verification

- verify patient identity using approved identifiers
- confirm procedure planned
- confirm laterality/site when applicable

- confirm consent is present and matches procedure

Clinical review

- review allergies
- review NPO status when relevant
- review anticoagulants and antiplatelets
- review renal function and contrast risk
- review pertinent history affecting access or procedure complexity
- review prior CABG, TAVR, peripheral intervention, or device history if relevant

Baseline clinical status

- review baseline blood pressure, heart rate, SpO2, and rhythm
- assess pulses and limb status when relevant
- assess pain and symptom status
- assess airway if sedation is planned
- confirm IV access or other required access is available

Safety readiness

- emergency equipment immediately available
- defibrillator and pads available when indicated
- oxygen setup available
- suction available if needed
- medications available for routine and emergency use

Equipment Readiness Checklist

- monitors connected and functioning
- pressure transducer zeroed and waveform ready
- manifold setup complete
- contrast available
- nitroglycerin available when needed
- ACT supplies available
- procedural wires, catheters, sheaths, and closure supplies available
- bailout supplies locatable immediately
- shielding in place
- imaging equipment functional

Epic Cupid Pre-Procedure Workflow

The following should be reviewed or completed before case start as applicable: - Patient in Room - Patient ID - Verifications and Consents - Oxygen - Pulse Ox Placement - Basic Vitals - ECG Rhythm - Pulses - Pain Assessment - Airway Assessment Complete - Site Preparation - Pressure and Skin Injury Prevention - Grounding Pads / Defib Pads if indicated - Fire Risk Assessment - Pre-Procedure Checklist - Physician Paged/Communication when relevant - MD pre-sedation check when applicable

Pre-Procedure Room Readiness Check

Before the operator is ready to begin, confirm: - correct procedure listed in the panel - staff assigned correctly in Epic Cupid - patient monitoring active - medications prepared - access site prepared - timeout readiness complete - documentation not missing critical baseline elements

Red Flags Before Starting

Do not proceed without addressing: - identity mismatch - missing or incorrect consent - no baseline vitals or rhythm documented - no pulse assessment when clinically relevant - missing airway assessment before sedation - missing access supplies - no clear access plan - no available emergency equipment - incomplete site preparation - unresolved allergy or contrast concern

Pre-Procedure Checklist for the Binder

Minimum before timeout: - patient identified - procedure confirmed - consent verified - allergies reviewed - access plan known - baseline vitals complete - rhythm documented - oxygen plan confirmed - emergency equipment available - sedation readiness complete if applicable - monitoring active - documentation current

Timeout Elements

The timeout should verify at minimum: - correct patient - correct procedure - correct side/site when relevant - allergies reviewed - anticoagulation status reviewed - sedation plan reviewed - access plan reviewed - required equipment available - team agreement to proceed

Preceptor Teaching Points

Preceptors should reinforce that pre-procedure preparation is not an administrative delay. It is the phase where preventable safety failures are most often avoided.

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4. Intra-Procedure Monitoring and Safety

Purpose

Train staff to monitor the patient, equipment, and procedure actively from start to finish. Monitoring is continuous interpretation, not passive observation.

Core Principle

The patient’s condition, the waveform, the ECG, the access site, the equipment, and the procedural environment can all change within seconds. Staff must know what is normal, what is abnormal, and what requires immediate escalation.

Continuous Monitoring Priorities

During every case, maintain active awareness of: - patient symptoms and level of responsiveness - blood pressure and arterial waveform - heart rate and rhythm - oxygen saturation and respiratory status - sedation effect - contrast volume - radiation exposure - ACT timing and results when applicable - access site integrity - sterile field integrity - catheter, wire, and device behavior

Patient Monitoring

Observe continuously for: - chest pain - shortness of breath - diaphoresis - agitation - nausea or vasovagal symptoms - reduced responsiveness - oversedation or respiratory depression

ECG and Rhythm Monitoring

Watch for: - new ST elevation - new ST depression - bradycardia - tachyarrhythmia - ventricular ectopy - wide-complex rhythm change - pauses or conduction delay

Immediate reporting is required for any new ischemic or unstable rhythm change.

Pressure Waveform Monitoring

Watch for: - hypotension - severe hypertension - dampened waveform - ventricularization - sudden waveform loss or distortion

A dampened waveform must be treated as urgent until proven otherwise.

Oxygenation and Respiratory Monitoring

Observe for: - falling SpO2 - shallow respirations - reduced responsiveness after sedation - airway obstruction signs - increased oxygen requirement

Access Site Monitoring

Monitor for: - active bleeding - enlarging hematoma - swelling - pain out of proportion - loss of distal perfusion concern - sheath instability or dislodgement

Contrast Awareness

The assigned team member should maintain awareness of total contrast used throughout the case and call out rising totals when clinically relevant.

Radiation Awareness

Radiation exposure should be tracked throughout the case. Staff should maintain shielding, detector positioning, and dose awareness rather than waiting until the end of the procedure to review exposure.

Medication and Sedation Safety

During the case, staff should maintain awareness of: - medication timing - total sedative effect - blood pressure response to medication - respiratory response to sedation - reversal or rescue needs - charting completion in Epic Cupid

Device and Equipment Safety

Call out and address immediately: - resistance during wire advancement - resistance during device delivery - unexplained catheter instability - equipment malfunction - break in sterile workflow - missing or incorrect supply entry if it affects the procedure

Escalation Triggers

Immediate verbal escalation is required for: - new ischemic symptoms - sudden hypotension - dampened pressure waveform - new ST change - unstable arrhythmia - oxygen desaturation - oversedation - access bleeding - enlarging hematoma - suspected perforation - suspected no-reflow - suspected dissection - unexpected device resistance or entrapment - sterile breach

Standard Call-Out Language

The orientee should be trained to use short direct call-outs such as: - “Pressure dampening.” - “New ST changes.” - “Blood pressure dropping.” - “SpO2 falling.” - “Site bleeding.” - “ACT due.” - “Contrast increasing.” - “Radiation rising.” - “Resistance advancing device.”

Epic Cupid Monitoring-Related Workflow Areas

Monitoring-related documentation may involve: - Basic Vitals - ECG Rhythm - Pulses - Pain Assessment - Sedation Assessment - Meds / MAR - POCT - Hemodynamic Monitoring entries in the Event Log - Radiation Tracking - Contrast Injected & Used - Site condition

Common Monitoring Errors

These should be reviewed during orientation: - watching the screen without interpreting the data - noticing a change but not calling it out promptly - focusing on one task and losing awareness of patient status - charting late and reconstructing events from memory - failing to reassess after medication administration - failing to reassess after access, intervention, or closure steps

Intra-Procedure Safety Pause

When the case becomes unstable or unexpectedly complex, staff should mentally reset and identify: - patient status - hemodynamic status - ECG status - access status - device position - medication/sedation status - immediate next step

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5. Coronary Angiography Fundamentals

Purpose

Teach new staff the basic logic, sequence, and safety principles of diagnostic coronary angiography so they can follow the case accurately, anticipate procedural needs, and recognize unsafe changes early.

Core Principle

Angiography is not simply injecting contrast and recording images. It is a stepwise process that depends on safe catheter position, accurate waveform interpretation, deliberate imaging, and constant reassessment of the patient and coronary flow.

Orientation Goal

By the end of this section, the orientee should be able to: - describe the standard sequence of coronary angiography - explain why waveform assessment matters before injection - recognize the difference between safe engagement and unsafe engagement - identify the major findings the operator is evaluating on each run - recognize common angiography-related complications and warning signs

Standard Angiography Sequence

1. Confirm safe catheter position

Before contrast injection, confirm that the catheter is appropriately engaged and the pressure waveform is acceptable. A poor waveform, deep seating, or noncoaxial position increases the risk of dissection, ischemia, and poor imaging.

2. Reassess pressure waveform

The team should be alert for: - dampening - ventricularization - sudden loss of normal waveform contour - unstable blood pressure response

If the waveform is not appropriate, the issue should be addressed before injection.

3. Use a test injection when needed

A small test injection may help confirm position and vessel opacification before a full cine run, especially when engagement is uncertain.

4. Administer intracoronary nitroglycerin when appropriate

Nitroglycerin may be used to reduce spasm and improve vessel sizing when clinically appropriate. Staff should understand that nitroglycerin affects interpretation and patient hemodynamics, so blood pressure awareness remains important.

5. Optimize imaging position

Image acquisition should aim for: - minimal vessel overlap - minimal foreshortening - adequate distal vessel visualization - good detector positioning - appropriate shielding and radiation awareness

6. Perform cineangiography

During the run, staff should monitor: - patient symptoms - ECG changes - pressure changes - catheter stability - contrast flow quality - operator requests for equipment or repositioning

7. Reassess immediately after the run

After each run, reassess: - patient status - ECG - waveform - blood pressure - evidence of spasm, dissection, embolization, or instability

8. Interpret the major findings

Although the final interpretation belongs to the operator, the orientee should understand that angiography is evaluating: - vessel patency - TIMI flow - lesion severity - lesion length - calcification - thrombus burden - branch involvement - dissection - tortuosity - ostial involvement - distal runoff and collateralization when relevant

What the Orientee Should Watch During Angiography

The orientee should not look only at the monitor screen. They should actively track all of the following: - pressure waveform before injection - ECG during engagement and injection - patient symptoms during the run - contrast volume accumulation - radiation accumulation - catheter stability - whether the vessel fills briskly or poorly - whether flow appears slow, interrupted, or threatened after intervention

Common Angiography Views and Concepts

The binder should later include a quick-reference page for common coronary views, but at the orientation stage the key concept is that multiple views are required because one image can misrepresent lesion severity due to overlap or foreshortening.

Angiography-Related Red Flags

Immediate concern is warranted for: - pressure dampening before or during injection - sudden chest pain with injection - new ST changes after engagement or injection - poor coronary filling - abrupt vessel cutoff - visible dissection flap - air or thrombus appearance - hypotension after injection - refractory spasm

Common Angiography Problems

Coronary dissection

May follow unsafe engagement or injection during dampening.

Air embolization

Can occur if the system is not properly cleared or managed.

Thrombus or debris embolization

Can impair distal flow and cause ischemia.

Spasm

May mimic or worsen obstructive disease and cause symptoms or ECG changes.

Poor image acquisition

Overlap, foreshortening, motion, or incomplete filling can limit interpretation.

Staff Responsibilities During Angiography

RN

- monitor symptoms and hemodynamics
- respond to medication requests
- track sedation effect
- identify and escalate ischemic or hemodynamic change
- maintain charting awareness

Radiology Tech

- optimize imaging support
- assist with catheter and equipment readiness
- maintain radiation awareness
- anticipate the next imaging or device need
- protect sterile workflow

Epic Cupid Relevance

Angiography-related documentation may involve: - Diagnostic Heart Cath - Diagnostic Peripheral Cath - Event Log - Radiation Tracking - Contrast Injected & Used

Suggested Preceptor Focus for This Tab

Early teaching should emphasize safe engagement, waveform awareness, and what to watch before and after each injection. The orientee does not need advanced lesion interpretation at this stage.

Common Angiography Workflow Errors

- injecting during dampening or unsafe engagement
- focusing only on the image and missing the patient response
- failing to reassess pressure or ECG after a run
- under-recognizing spasm, dissection, or poor distal flow

- poor awareness of contrast and radiation accumulation during repeated runs

Teaching Pearls

- A normal-looking run in one view does not exclude significant disease.
- Pressure waveform quality matters before every injection.
- A patient complaint during injection matters, even if the image looks acceptable.
- Slow flow after intervention or angiography should never be dismissed.

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6. Medications in the Cath Lab

Purpose

Provide a practical Cath Lab RN medication framework tied to monitoring, documentation, escalation, and postprocedure safety.

Core Principle

In the Cath Lab, medications change what the RN must watch next. The bedside nurse should link every medication to the immediate monitoring priority, expected effect, and likely adverse effect.

Orientation Goal

By the end of this section, the orientee should be able to identify the major medication groups used in the Cath Lab, understand why a medication is being given in the moment, recognize what monitoring changes after administration, identify high-risk adverse effects that require escalation, and document medication use accurately in Epic Cupid and MAR.

Sedation and Reversal

Midazolam (Versed)

Typical RN use: anxiolysis and procedural sedation support. Typical dosing in the local medication guide: 0.5-1 mg IV, repeat as needed. Duration: 15-80 minutes. Primary RN concern: hypopnea or respiratory depression. If significant respiratory compromise occurs, stop additional sedation, support oxygenation and ventilation, and escalate immediately. Reversal is flumazenil 0.2 mg IV over 15 seconds, then repeat 0.2 mg after 45 seconds and every 1 minute as needed to a maximum of 1 mg.

Fentanyl

Typical RN use: analgesia during painful portions of the case. Typical dosing in the local medication guide: 25-100 mcg IV, repeat as needed. Duration: 30-60 minutes. Primary RN concerns: respiratory depression, oversedation, and delayed gastric emptying, which can delay oral P2Y12 absorption. Avoid opioids in STEMI when not necessary if delayed oral antiplatelet

effect is a concern. Reversal is naloxone 0.1-0.2 mg IV, repeated every 2-3 minutes until desired reversal.

OLG procedural sedation standards relevant to Cath Lab RN practice

At Ochsner Lafayette General, moderate procedural sedation requires one physician credentialed in PSA and one RN with completed competencies. RNs may monitor and administer medications for moderate sedation. For deep sedation, RNs may monitor only and may not administer the deep-sedation medications. The RN may not leave the patient unattended until the sedation protocol and recovery phase are complete. Required monitoring includes pulse oximetry, ECG or cardiac rate and regularity, blood pressure, respiratory rate and quality, level of consciousness, pain as appropriate, and continuous ventilation assessment by observation and capnography. Recovery documentation is performed at intervals no greater than every 15 minutes until recovery criteria are met, and monitoring continues for a minimum of 30 minutes after the last sedating medication or 30 minutes after reversal if reversal was used.

Coronary Vasodilators / No-Reflow / Spasm

Nitroglycerin

Typical RN relevance: coronary spasm treatment or prevention and IV anti-ischemic support. Local guide: infusion starts at 10 mcg/min and increases by 10 mcg/min every 5 minutes while maintaining systolic blood pressure above 100. Maximum 200 mcg/min. Sublingual dose 0.4 mg. Avoid in hypotension, RV infarct, recent PDE-5 inhibitor use, and hypertrophic obstructive cardiomyopathy. If hypotension occurs, stop additional doses, give normal saline, wait, and escalate to vasopressor support if persistent.

Nicardipine, Nitroprusside, Verapamil, Adenosine

The local Cath Lab guide lists these as common agents for no-reflow and spasm management. Nicardipine 100-300 mcg IC and nitroprusside 100-300 mcg IC are listed for prevention or treatment of no-reflow, with nicardipine noted as preferred versus nitroprusside or verapamil because of less hypotension and shorter action. Verapamil is listed for radial spasm prevention and no-reflow treatment, with radial dosing 2-3 mg and intracoronary dosing 1 mg over 2 minutes. Adenosine is listed for no-reflow treatment and physiologic testing, with example doses of 50-100 mcg IC for RCA, 100-200 mcg IC for left main, and higher slow IC doses for no-reflow. RN priorities after any of these include close blood pressure surveillance, rhythm monitoring, and rapid escalation for hypotension, AV block, AF, VF, or clinical deterioration.

Contrast / Allergy Management

The local guide emphasizes using the lowest feasible contrast volume and notes contrast-associated AKI risk, especially in advanced CKD. Premedication options listed include prednisone 50 mg at 13, 7, and 1 hour pre-procedure or methylprednisolone 32 mg at 12 and 2 hours pre-procedure, plus diphenhydramine 50 mg by mouth 1 hour pre-procedure. Treatment options listed include diphenhydramine 25-50 mg IV, cimetidine 300 mg IV or ranitidine 50 mg IV over 15 minutes, hydrocortisone sodium succinate 100-400 mg IV over 1 minute, and epinephrine 0.3 mg IV of 1:10,000 solution for anaphylactic shock, repeatable to 1 mg total.

Anticoagulation and ACT

Unfractionated heparin (UFH)

The local guide lists ACT checks every 20-30 minutes. Goal ACT without GP IIb/IIIa or cangrelor is Hemochron 300-350 seconds, with greater than 350 often recommended for retrograde CTO. Goal ACT with GP IIb/IIIa or cangrelor is 200-250 seconds. Protamine dosing for UFH reversal is 1-1.5 mg per 100 units of heparin, maximum 50 mg, given slowly. High-risk hypersensitivity factors listed include prior NPH or protamine insulin, fish hypersensitivity, vasectomy, severe LV dysfunction, and abnormal pulmonary hemodynamics.

Bivalirudin

Local guide dosing: 0.75 mg/kg IV bolus then 1.75 mg/kg/hour infusion, with renal infusion adjustment for lower creatinine clearance and dialysis. Half-life is about 25 minutes. Monitoring still uses ACT, with the same target framework as above, and there is no reversal agent, so stopping the infusion is the immediate response to bleeding concern.

Hemochron Signature Elite ACT handling points for RN orientation

Fresh whole blood in a non-heparinized syringe must be processed immediately. The cuvette must be at room temperature. Once the instrument displays Ready, the sample is applied and Start is pressed. Blood should fill the center sample well flush to the top, not be forced into the center pin, and air bubbles should be avoided. Results less than 65 seconds display Out of Range-Lo and should be repeated; results above 400 seconds display Out of Range-Hi; and any result inconsistent with the clinical picture should be repeated immediately.

Antiplatelets and IV Antiplatelet Support

The local guide includes aspirin 325 mg load then 81-100 mg daily; clopidogrel 600 mg load then 75 mg daily; ticagrelor 180 mg load then 90 mg twice daily; prasugrel 60 mg load then 10 mg daily, reduced to 5 mg daily if weight is under 60 kg or age is over 75; and cangrelor 30 mcg/kg IV bolus then 4 mcg/kg/min infusion for at least 2 hours or the duration of the procedure. For cangrelor transition, ticagrelor may be given anytime, whereas clopidogrel or prasugrel loading is done immediately after cangrelor is stopped. Eptifibatide and tirofiban infusion doses and renal reductions are also listed. The guide notes keeping ACT 200-250 seconds when IV antiplatelet support is used.

Vasopressors / Inotropes

The local guide includes phenylephrine 50-100 mcg IV bolus then 0.5-1.4 mcg/kg/min infusion, norepinephrine 0.1-0.5 mcg/kg/min infusion, vasopressin starting at 0.03 units/min with titration, dopamine 10-20 mcg/kg/min in vasopressor range, epinephrine 2-10 mcg/min infusion for cardiogenic shock, and dobutamine 10-40 mcg/kg/min. RN priorities are blood pressure trend, rhythm monitoring, access integrity, and fast reassessment after any titration.

Antiarrhythmics / Chronotropes

The local guide includes amiodarone 150 mg over 10 minutes, repeatable, followed by 1 mg/min for 6 hours, and atropine 0.5 mg IV push repeated to a maximum of 3 mg. Key RN risks listed for amiodarone include hypotension, bradycardia or AV block, and torsades or VF in prolonged QT.

High-Risk Medication Call-Outs for RN Orientation

Call out immediately for respiratory depression after midazolam or fentanyl; persistent hypotension after nitroglycerin, nicardipine, nitroprusside, or verapamil; AV block, AF, VF, or torsades after adenosine; unexpected bleeding during anticoagulation or IV antiplatelet use; ACT result inconsistent with therapy or clinical picture; protamine reaction with hypotension or pulmonary pressure concern; allergic reaction or anaphylaxis after contrast exposure; and bradycardia, hypotension, or rhythm instability after amiodarone or atropine-related response.

Epic Cupid Medication Workflow

Medication-related documentation may involve Meds / MAR, Sedation Assessment, Event Log medication entries, and POCT when medication timing relates to ACT or other testing.

Suggested Preceptor Focus for This Tab

Early medication teaching should focus on four questions: why is the medication being given, what should the RN watch next, where is it documented, and what finding would require immediate escalation.

Teaching Pearls

Every medication given in the Cath Lab should change the nurse's immediate monitoring priority. Sedation and antithrombotic medications create delayed risk if reassessment is skipped. In the RN role, medication safety is inseparable from real-time charting and recovery monitoring.

7. Access, Catheter Movement, and Closure

Purpose

Orient staff to access selection, access documentation, catheter movement safety, sheath workflow, and closure planning so they can support the case from vessel entry through postprocedure site management.

Core Principle

Access is not a single moment in the case. It is a continuous workflow that begins with access planning and ends only after hemostasis, site reassessment, and documentation are complete.

Orientation Goal

By the end of this section, the orientee should be able to: - describe the basic workflow for vascular access - recognize access-related safety risks - follow sheath, wire, and catheter movement conceptually - understand closure preparation and postprocedure site priorities - document access and closure workflow accurately in Epic Cupid

Access Planning Basics

Before access begins, the team should understand: - planned access site - reason for radial or femoral approach when relevant - sheath size expectations - whether larger devices or complex support may be needed - whether closure device use is anticipated - whether prior procedures or anatomy may affect access strategy

Standard Access Workflow

In your Epic Cupid build, access-related workflow includes: - Percutaneous Stick - Catheter Inserted - Sheath - Dilator - Wire - Evaluate for closure device - Sheath Exchange

This should be taught as a sequence rather than isolated charting tasks.

Access Safety Priorities

During access and sheath management, staff should monitor for: - bleeding - hematoma formation - unstable blood pressure - pain out of proportion - limb perfusion concerns - sheath instability or displacement - resistance during insertion or exchange

Catheter and Wire Movement Awareness

New staff do not need to perform the operator's role, but they must understand the safety implications of catheter and wire movement.

Important concepts include: - resistance matters - deep engagement can be dangerous - poor waveform during catheter position changes must be taken seriously - wire movement and device advancement should be tracked conceptually during the case - abrupt difficulty can signal tortuosity, calcification, dissection risk, or entrapment risk

Sheath Workflow

Sheath-related support includes: - correct sheath identification - size awareness - access-site laterality awareness - monitoring for bleeding or loss of integrity - documenting insertion, exchange, and removal events when applicable

Closure Planning

Closure should be anticipated before the end of the case, not improvised after it.

The team should know: - whether manual compression or a closure device is expected - whether anticoagulation status affects timing or method - whether postprocedure activity

restrictions will be different based on closure method - whether the site will require special monitoring

Post-Closure Monitoring Priorities

After closure or sheath removal, monitor for: - bleeding - swelling or hematoma - loss of distal perfusion concern - pain increase - hemodynamic instability - incomplete hemostasis

Epic Cupid Relevance

Access and closure-related charting may involve: - Access section - Postprocedure section - Closure Device Deployed - Incision Site - Site condition - All Catheters Removed - All Wires Removed - Hemostasis Time in Events tab when applicable

Suggested Preceptor Focus for This Tab

The orientee should first learn access as a sequence of events with specific safety checks rather than as isolated device names. Access-site vigilance should be reinforced in every case, not only during closure.

Common Access and Closure Errors

- access laterality omitted
- sheath size omitted
- wire or catheter events not documented
- closure device not documented
- site condition missing before transfer
- catheter or wire removal not documented
- hemostasis timing missing
- postprocedure pulse or site reassessment delayed or omitted

Teaching Pearls

- Access problems can declare themselves gradually or suddenly.
- Closure planning should begin before the last angiographic run is complete.
- The case is not finished when the intervention ends; it is finished when site stability and documentation are complete.

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8. Core PCI Workflow

Purpose

Give new staff a repeatable framework for PCI so they can anticipate the sequence of the case, prepare the correct equipment, and understand where safety checks belong.

Core Principle

PCI is easiest to follow when it is understood as a fixed series of phases. Even when the details change from case to case, the broad workflow remains recognizable.

Orientation Goal

By the end of this section, the orientee should be able to: - name the standard phases of a PCI case - anticipate what usually comes next in routine intervention - recognize when the case is leaving routine workflow - support equipment flow, monitoring, and charting through the procedure

Standard PCI Sequence

1. Planning
2. Monitoring readiness
3. Medication readiness
4. Access
5. Guide engagement
6. Angiography and target lesion review
7. Wiring
8. Lesion preparation
9. Physiology or imaging as needed
10. Stent delivery and deployment
11. Post-dilation or optimization as needed
12. Final angiographic assessment
13. Closure and postprocedure transition

Phase 1: Planning

Before PCI begins, the team should know: - target lesion or vessel - access plan - basic equipment expectations - anticoagulation strategy - whether imaging or physiology is likely - whether lesion complexity suggests a higher-risk case

Phase 2: Monitoring and Medication Readiness

Before the intervention advances, confirm: - monitoring is active and visible - hemodynamic baseline understood - sedation plan active if needed - anticoagulation awareness maintained - key medications available

Phase 3: Access and Guide Engagement

Once access is obtained, the team should support: - stable access workflow - appropriate guide or catheter readiness - awareness of waveform changes during engagement - rapid response to dampening or unstable hemodynamics

Phase 4: Wiring

Wiring is a major procedural transition. At this point, staff should become more alert to: - ischemic changes - rhythm changes - patient symptoms - device requests coming faster - the need for additional balloons, imaging tools, or support devices

Phase 5: Lesion Preparation

Lesion preparation may include one or more preparation steps before stent delivery.

Teaching priorities: - preparation matters for deliverability and expansion - more difficult lesions often require more deliberate preparation - the team should anticipate escalating equipment complexity if the lesion is calcified, long, or anatomically challenging

Phase 6: Physiology or Intravascular Imaging

These tools may be used to: - confirm lesion significance - clarify anatomy - size the vessel - guide stent length or diameter decisions - evaluate the final result

The orientee should understand their purpose even if they are not directly manipulating the device.

Phase 7: Stent Delivery and Deployment

During stent delivery and deployment, the team should be especially attentive to: - hemodynamic instability - ischemic ECG changes - pressure waveform changes - difficulty delivering equipment - side-branch compromise concern - immediate post-deployment flow result

Phase 8: Final Assessment

Before moving to closure, confirm that the team has considered: - flow result - branch patency - residual stenosis concern - dissection concern - no-reflow or distal embolization concern - final imaging or physiology if used - access and closure readiness

Phase 9: Closure and Transition

The PCI workflow is not complete until: - site and closure plan are clear - postprocedure orders and expectations are known - charting is complete enough for safe transfer - handoff includes relevant post-PCI concerns

Signs the Case Is Becoming More Complex

The orientee should recognize that the case may be moving out of routine workflow when there is: - repeated equipment exchange - difficulty wiring the lesion - escalating guide or support needs - calcification or tortuosity complicating deliverability - prolonged inflation or repeated preparation steps - new instability, slow flow, or branch compromise

Epic Cupid Relevance

PCI-related documentation may involve: - Diagnostic Heart Cath - Access - Meds / MAR - POCT - Supplies and Implants - Radiation Tracking - Contrast Injected & Used - Site condition - Document Case End

Suggested Preceptor Focus for This Tab

The best early teaching approach is to keep bringing the orientee back to the current PCI phase. Once they can identify the phase reliably, anticipation and charting accuracy improve significantly.

Common PCI Workflow Errors

- losing awareness of ACT timing during a long case
- focusing on equipment and missing patient instability
- failing to recognize the case has become high risk
- delayed documentation of key device use or case phases
- incomplete postprocedure transition planning

Teaching Pearls

- The easiest way to follow PCI is to know what phase the case is in.
- A routine case can become unstable quickly.
- The best support staff are already preparing the next likely need before the operator asks.
- Final result assessment is part of the procedure, not an afterthought.

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9. Complex Lesions and High-Risk Cases

Purpose

Prepare new staff to recognize when a case is higher risk, more technically demanding, or more likely to require nonroutine equipment, prolonged workflow, or rapid escalation.

Core Principle

Complex cases are safer when the team recognizes complexity early. The orientee does not need to master every advanced strategy initially, but should understand why certain cases demand more caution, more preparation, and more vigilance.

Orientation Goal

By the end of this section, the orientee should be able to: - identify common features that make a case high risk - recognize when a routine case is becoming more complex - anticipate changes in setup, equipment, and monitoring needs - understand the major complication patterns

associated with common complex lesion subsets - support the team without losing awareness of the overall procedural phase

General Features of a High-Risk Case

Cases are more likely to be high risk when they involve: - left main disease - bifurcation disease - heavy calcification - thrombus burden - hemodynamic instability - prior CABG or graft intervention - TAVR reaccess - long lesions or diffuse disease - small or very large vessels - tortuous anatomy - SCAD or fragile vessel pathology - need for prolonged procedure time or multiple device exchanges

What Changes in a High-Risk Case

Compared with routine cases, high-risk cases often require: - more detailed planning - more deliberate access decisions - increased equipment readiness - stronger anticipation of bailout needs - closer hemodynamic and ECG surveillance - more contrast and radiation awareness - more precise charting of procedural steps and case timing

Left Main PCI

Why it is high risk

Left main disease places a large amount of myocardium at risk. Balloon inflation, guide instability, poor expansion, or abrupt closure can cause rapid instability.

Staff preparation priorities

- anticipate a more deliberate setup
- ensure hemodynamic monitoring is closely followed
- be ready for imaging tools and additional lesion preparation devices
- expect strong attention to final optimization

Key concerns

- hemodynamic instability
- inadequate lesion preparation
- underexpansion
- side-branch compromise in distal left main bifurcation disease

Bifurcation Disease

Why it is high risk

Bifurcation PCI adds the risk of side-branch compromise, rewiring complexity, extra device exchanges, and longer procedure time.

Staff preparation priorities

- anticipate more than one wire
- expect additional balloon work
- be alert for side-branch loss or changing flow

- follow the operator's sequence closely because equipment use can escalate quickly

Key concerns

- side-branch compromise
- prolonged case time
- need for kissing inflations or additional stenting
- increased contrast and radiation use

Calcified Lesions

Why they are high risk

Calcified lesions can resist balloon expansion, impair stent delivery, and increase the chance of underexpansion or dissection.

Staff preparation priorities

- expect more deliberate lesion preparation
- anticipate need for specialty equipment
- recognize that the case may progress more slowly with higher procedural intensity

Key concerns

- difficult device delivery
- stent underexpansion
- prolonged procedure time
- increased complication risk during aggressive preparation

Thrombotic ACS or STEMI Cases

Why they are high risk

These cases combine unstable coronary pathology with time sensitivity and high risk of abrupt flow compromise, distal embolization, no-reflow, arrhythmia, and hemodynamic deterioration.

Staff preparation priorities

- maintain urgency without losing control of workflow
- track symptoms, ECG, and hemodynamics closely
- anticipate additional medication use and rapid procedural changes
- maintain strong awareness of reperfusion timing and case-phase timing

Key concerns

- no-reflow
- distal embolization
- unstable rhythm
- hypotension or shock
- recurrent ischemia

Graft Cases

Why they are high risk

Graft intervention can involve friable plaque, embolization risk, challenging anatomy, and nonroutine imaging or device needs.

Staff preparation priorities

- anticipate a less routine equipment sequence
- maintain close attention to flow changes and hemodynamic instability
- be prepared for longer or more technically variable workflow

Key concerns

- embolization
- poor distal flow
- unstable hemodynamics
- prolonged intervention

TAVR Reaccess Cases

Why they are high risk

Coronary reaccess after TAVR may be more technically challenging due to altered anatomy and engagement complexity.

Staff preparation priorities

- anticipate more time for engagement
- maintain close pressure and ECG awareness during catheter positioning
- prepare for the possibility of more complex access or catheter strategy

Key concerns

- difficult engagement
- prolonged fluoroscopy time
- guide instability
- increased technical complexity early in the case

SCAD Awareness

Why it is high risk

SCAD involves fragile vessel pathology and can worsen with additional manipulation. These cases require restraint, precision, and careful recognition of changing anatomy.

Staff preparation priorities

- understand that the vessel may behave differently than routine atherosclerotic disease
- track symptoms and instability closely
- avoid normalizing repeated difficulty or abrupt anatomy change

Key concerns

- propagation of dissection
- loss of flow
- worsening ischemia
- increased risk from repeated instrumentation

Long Lesions and Diffuse Disease

Why they are high risk

These lesions often increase procedure length, device count, stent planning complexity, and risk of under-treatment or underexpansion.

Staff preparation priorities

- anticipate more devices and more contrast use
- follow inventory and charting carefully
- maintain awareness that the case may appear routine but become logistically complex

Key concerns

- high device burden
- long fluoroscopy time
- high contrast use
- more complex postdeployment assessment

Small and Large Vessel Challenges

Small vessels

These may be more sensitive to spasm, more difficult to size, and more prone to appearance changes with vasodilator use.

Large vessels

These may require more careful imaging and sizing awareness to avoid incomplete treatment or poor final optimization.

Hemodynamic Support and Shock Cases

Why they are high risk

These cases have limited physiologic reserve and can deteriorate quickly.

Staff preparation priorities

- maintain constant monitoring awareness
- anticipate rapid medication, equipment, and documentation changes
- prepare for multiple simultaneous priorities

Key concerns

- hypotension
- arrhythmia
- escalating support needs
- prolonged case duration

Signs the Case Is Escalating in Real Time

The orientee should recognize escalation when there is: - repeated guide, wire, or catheter change - unexpected difficulty crossing or delivering equipment - prolonged lesion preparation - recurrent hemodynamic instability - repeated ECG change - increased medication intensity - increasing urgency in operator requests - talk of bailout or additional support

Staff Responsibilities in Complex Cases

RN

- maintain close surveillance of symptoms, rhythm, blood pressure, and sedation effect
- anticipate medication and rescue needs
- maintain charting discipline even as the case accelerates
- communicate instability immediately

Radiology Tech

- anticipate additional device and imaging needs
- maintain room organization during escalating complexity
- protect sterile workflow under pressure
- help the operator move efficiently through equipment changes

Teaching Pearls

- Complexity should be recognized early, not after the case is already unstable.
- The orientee's job in a complex case is not to know every advanced maneuver; it is to remain organized, observant, and anticipatory.
- High-risk cases punish poor communication and delayed recognition.

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10. Complications and Bailout Algorithms

Purpose

Provide a structured framework for recognizing procedural complications quickly, escalating them immediately, and supporting bailout workflow safely.

Core Principle

The first step in managing any complication is recognition. The second is clear escalation. The third is organized support. Staff should not wait to be completely certain before reporting a dangerous change.

Orientation Goal

By the end of this section, the orientee should be able to: - recognize common complication patterns in the Cath Lab - identify the first actions that support safe response - communicate the problem clearly and early - maintain role discipline during a high-stress event - understand the purpose of bailout tools and emergent workflow changes

Universal Response Framework

When a complication is suspected: 1. Recognize the change 2. Call it out clearly 3. Reassess patient status, hemodynamics, ECG, and access 4. Stay organized and prepare for rapidly changing requests 5. Chart key events in real time when feasible 6. Support transition to bailout workflow, recovery, or emergency response

Acute Vessel Closure

Warning signs

- abrupt flow loss
- worsening chest pain
- ischemic ECG change
- hemodynamic instability
- sudden angiographic change

First support priorities

- maintain awareness of whether wire position is still preserved
- prepare for rapid device exchange and urgent escalation
- monitor symptoms, ECG, and blood pressure continuously
- document key timing and major events as able

Key concept

If the wire is already across the lesion, preserving wire position is often critical.

Coronary Dissection

Warning signs

- pressure dampening
- abrupt chest pain
- sudden angiographic flap or staining
- loss of flow
- ischemic ECG change

- instability after engagement, ballooning, or injection

Support priorities

- stop treating the case like routine angiography or routine PCI
- anticipate rapid change in device needs
- maintain close hemodynamic monitoring
- be ready for urgent escalation in treatment intensity

No-Reflow or Distal Embolization

Warning signs

- slow or poor distal flow after intervention
- recurrent chest pain
- ST change
- hypotension
- worsening rhythm instability

Support priorities

- recognize that the artery may be open but perfusion still poor
- anticipate additional medication use
- track blood pressure and rhythm closely
- maintain rapid communication and charting awareness

Perforation

Warning signs

- sudden hemodynamic instability
- angiographic extravasation
- chest pain or abrupt deterioration
- developing tamponade concern

Support priorities

- escalate immediately
- prepare for emergency workflow change
- watch blood pressure continuously
- anticipate emergent equipment and support needs

Air Embolism

Warning signs

- sudden chest pain
- abrupt ECG change
- hypotension
- angiographic evidence of air

- sudden hemodynamic collapse

Support priorities

- escalate immediately
- support rapid response to instability
- track rhythm and blood pressure without interruption

Side-Branch Occlusion

Warning signs

- change in branch flow after ballooning or stenting
- ischemic change
- recurrent chest pain
- angiographic side-branch compromise

Support priorities

- anticipate rewiring, ballooning, or additional stent strategy
- maintain close hemodynamic and symptom monitoring
- continue organized equipment support

Equipment Entrapment or Delivery Failure

Warning signs

- repeated resistance
- inability to advance or withdraw equipment
- increased urgency around device movement
- prolonged effort at one procedural step

Support priorities

- recognize that the case is no longer routine
- keep the room organized
- anticipate additional support devices or retrieval-related equipment
- monitor patient stability while the technical issue is addressed

Access-Site Bleeding or Hematoma

Warning signs

- visible bleeding
- swelling at access site
- pain increase
- blood pressure drop
- loss of site stability during or after sheath management

Support priorities

- call out the bleeding immediately

- reassess blood pressure and symptoms
- prepare for urgent site management
- document changes and actions clearly

Vasovagal Event

Warning signs

- bradycardia
- hypotension
- nausea
- diaphoresis
- pallor
- sudden patient discomfort during access or closure-related phases

Support priorities

- recognize the pattern quickly
- escalate immediately
- monitor blood pressure, rhythm, and responsiveness closely
- support medication and positioning needs promptly

Hemodynamic Collapse

Warning signs

- severe hypotension
- loss of perfusing rhythm
- altered responsiveness
- severe ischemia
- sudden escalation of team urgency

Support priorities

- switch mentally from routine procedure support to emergency support
- maintain role clarity
- anticipate code or shock workflow
- keep documentation, timing, and communication as organized as possible

Code Workflow

When the case transitions into a code or major emergency: - use the dedicated Code workflow in Epic Cupid as appropriate - prioritize call-outs, timing, and medication/event awareness - avoid relying on memory after the fact for critical emergency events

Documentation During Complications

Even in unstable events, staff should try to preserve documentation of: - time of major change - hemodynamic instability - rhythm change - medications given - major procedural events - transition to recovery, emergency support, or procedure end

Common Complication Management Errors

- delayed recognition
- delayed call-out because the staff member is uncertain
- focusing on one device or one screen and missing the patient's condition
- losing charting discipline entirely during a prolonged unstable event
- poor handoff after stabilization

Teaching Pearls

- Most complications announce themselves before they fully declare themselves.
- Early call-outs save time.
- Staying organized during a complication is a clinical skill.
- The team does better when every member remains within role and communicates clearly.

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11. Epic Cupid Procedure Charting

Purpose

Teach new Cath Lab nurses and radiology techs how to chart procedures in Epic Cupid using the actual local Procedure Log workflow, left-side toolbox structure, timing logic, and postprocedure sequence used in this lab.

Core Principle

In this build, the left panel is the workflow entry system, the Event Log is the legal chronological story of the case, and the Events tab is the master timing control for major case phases. Staff should chart in the sequence the case unfolds and confirm that key entries appear in the correct workflow areas.

Orientation Goal

By the end of this section, the orientee should be able to: - navigate the Procedure Log confidently - identify where pre-procedure, intra-procedure, and postprocedure charting belongs - complete routine case charting in real time with minimal prompting - identify missing documentation before the patient leaves Cath Lab care - understand how procedure timing, supplies, access, sedation, and recovery documentation connect to one another

Procedure Log Overview

Main components of the screen

- top Epic navigation
- Procedure Log tabs
- left workflow toolbox
- central Event Log
- structured timing and postprocedure sections

Procedure Log tabs visible in your build

- Event Log
- Staff
- Procedures
- Supplies and Implants
- Events
- Timeout
- Labs/Orders
- Counts
- Charge Capture
- Patient Positioning

Training rule

The orientee should chart in real time whenever possible. The case should not be reconstructed from memory after transfer.

Left Toolbox Workflow Map

Procedure / Physicians / Panel

Used to verify the case framework, physician assignment, and panel details. Procedure-end items may also appear here depending on workflow.

Staff

Used to assign or verify staff members and room roles.

Head-to-Toe Assessment

Used for general nursing assessment.

Procedure Assessment and Prep

This is the front-end readiness section. Visible items include: - Patient in Room - Patient ID - Verifications and Consents - Aldrete Score - Oxygen - Pulse Ox Placement - Basic Vitals - ECG Rhythm - Pulses - Pain Assessment - Airway Assessment Complete - Site Preparation - Pressure and Skin Injury Prevention - Grounding Pads - Defib Pads - 3D Mapping Patch - EP Patient Prep

- EP Procedure Device Programming - Fire Risk Assessment - Pre-Procedure Checklist - Physician Paged/Communication - MD pre-sedation check

Document Case Start

Used for formal case-start workflow.

Sedation Assessment

Used for sedation-related documentation.

Add an LDA / Existing LDAs / PICC Line

Used for line and device documentation.

Meds / MAR

Used for medication administration.

POCT

Used for point-of-care testing such as ACT when applicable.

Access

Used to chart: - Percutaneous Stick - Catheter Inserted - Sheath - Dilator - Wire - Evaluate for closure device - Sheath Exchange

Diagnostic Heart Cath

Visible workflow items include: - Wire - Catheter - Angiography - Specimen - Intracoronary Vasodilator - FFR/iFR

TEE/Cardioversion

Used when those workflows apply.

Diagnostic Peripheral Cath

Visible workflow items include: - Wire - Catheter - Angiography - Specimen - Intra-arterial Vasodilator - Peripheral Pressure - Angiographic Measurement

Peripheral Upper Venous / Lower Extremity / Intervention

Used for venous and peripheral intervention workflows. The peripheral intervention section includes a broad set of interventional entries such as balloons, atherectomy, thrombectomy, laser, stents, drug-coated balloon, closure, thrombolytic infusion, and other specialized treatment steps.

Sheath Removal

Used when sheath removal is charted as a distinct workflow.

Postprocedure

Visible workflow items include: - In Cath Recovery - Radiation Tracking - Contrast Injected & Used - Site condition - Arm Sling - Pain Assessment - Basic Vitals - Pulses - ECG Rhythm - Aldrete Score - Closure Device Deployed - Incision Site - All Catheters Removed - All Wires Removed - Out of Cath Recovery

Document Case End

Used for: - Case End - Sedation End

Patient Movement

Used to document movement or transition in care.

Care Plan/Education

Used for patient education and care planning.

Misc.

Used for additional workflow items.

Code

Used for emergency/code workflow such as: - CPR Start - Code Narrator/Flowsheet - Medication - Airway - Rhythm - Arrival - CPR Ended - other code-related interventions

Standard Chronological Charting Workflow

Phase 1: Pre-procedure

Complete or verify: - procedure/panel - staff assignment - patient in room - patient ID - verifications and consents - oxygen and pulse ox placement - basic vitals - ECG rhythm - pulses - pain assessment - airway assessment if sedation planned - site preparation - fire risk assessment - pre-procedure checklist - MD pre-sedation check when applicable

Phase 2: Case start

Document: - timeout - case start - procedure start - sedation start when applicable

Phase 3: Intra-procedure

Local Case Tracking Rules

Use the Events tab as the official source of truth for case-phase timestamps. In this local build, several events are operationally important because they drive room-time and recovery-time workflow.

Required event expectations

In Pre-Procedure and Pre-Procedure Complete are manually selected by nursing and are part of the verify workflow. In Room must be documented when the patient is physically rolled into the Cath Lab. Case Start (Cath Only) should reflect that the provider is in the room, ready to begin, and the pre-incision timeout is complete. Case End (Cath Only) should reflect that the case is complete and debrief has been performed. Out of Room must be documented when the patient physically leaves the suite. If the procedure is not performed, use Procedure Aborted.

Document: - medications - POCT - access sequence - catheter/wire/sheath workflow - angiography - procedural devices and interventions - hemodynamic changes - supplies and implants - major complications and workflow changes

Phase 4: Postprocedure / room completion

Document: - radiation tracking - contrast injected and used - site condition - closure device deployed if applicable - incision site - all catheters removed - all wires removed - pain reassessment - vitals - pulses - ECG rhythm - Aldrete score when applicable

Phase 5: Formal closeout

Document: - case end - sedation end - patient movement - recovery progression - handoff completion

Events Tab Workflow

Purpose

The Events tab is the official timing control for the case lifecycle.

Event groups visible in your build

Pre-proc

- In Facility
- In Pre-Procedure
- Pre-Procedure Complete
- Out of Pre-Procedure
- Holding Start
- Holding Stop

Intra-proc

- In Room
- Sedation Start
- Procedure Start
- Case Start (Cath Only)
- STEMI reperfusion criteria met
- Case End (Cath Only)
- Procedure Finish

- Sedation End
- Hemostasis Time
- Out of Room

Panel-specific timing

Each panel procedure can have its own start and end time.

Post-proc

Visible items include recovery phases such as: - In Cath Recovery - Out Cath Recovery - In Recovery - Recovery Care Complete - Out of Recovery - In Phase II - Phase II Care Complete - Out of Phase II - Procedural Care Complete - Procedure Aborted - and other recovery-related states

Anesthesia

Visible items include: - Anesthesia Start - Anesthesia Finish - Emergence - related anesthesia timing fields

Training standard

These timestamps should be entered in real time or immediately when the event occurs. They should not be batch-entered later from memory.

Timing concepts that must be understood

- Procedure Start is not automatically the same as Case Start.
- Case End is not automatically the same as Procedure Finish.
- Sedation End is not automatically the same as Out of Room.
- Hemostasis Time may occur after Case End.
- Recovery phases should reflect the patient's actual progression.

Timeout Workflow

Timeout categories visible in your build

- Preprocedure
- Pre-incision
- Debriefing

Training standard

Timeout should be completed in real time and should reflect actual procedural pause and verbal confirmation, not retrospective completion.

Minimum timeout expectations

Before access or intervention, confirm: - correct patient - correct procedure - correct site/side when relevant - consent present - allergies reviewed - access plan reviewed - sedation plan reviewed when applicable - key equipment available

At debriefing, confirm: - procedure performed - major devices or implants documented - postprocedure plan - recovery destination - immediate concerns to monitor

Supplies and Implants Workflow

Purpose

Document supplies and implants used during the case.

Key training points

- chart supplies during the case whenever possible
- verify quantity used
- verify wasted quantity when applicable
- distinguish routine supply from implant
- correct missed entries before the patient leaves Cath Lab care

Common errors

- omitted device entry
- incorrect quantity
- missed wasted item documentation
- implant not identified properly

Access Documentation Workflow

Purpose

To chart vascular entry and related procedural access events in sequence.

Standard sequence in your build

- Percutaneous Stick
- Catheter Inserted
- Sheath
- Dilator
- Wire
- Evaluate for closure device
- Sheath Exchange

Required details to verify

- side or laterality
- site
- vessel when applicable
- device type
- sheath size
- exchange events
- closure planning

Postprocedure Workflow

Purpose

Document immediate recovery and completion of procedure-related tasks.

Postprocedure items visible in your build

- In Cath Recovery
- Radiation Tracking
- Contrast Injected & Used
- Site condition
- Arm Sling
- Pain Assessment
- Basic Vitals
- Pulses
- ECG Rhythm
- Aldrete Score
- Closure Device Deployed
- Incision Site
- All Catheters Removed
- All Wires Removed
- Out of Cath Recovery

Training standard

The case is not complete until postprocedure recovery and removal-related documentation are complete.

Common Charting Errors

- missing timeout
- missing patient ID or consent verification
- incomplete baseline vitals before case start
- sedation start or sedation end missing
- access laterality or sheath size omitted
- supplies not entered
- contrast or radiation not documented
- site condition missing
- all wires or catheters removed not documented
- case end omitted
- patient movement or handoff incomplete
- major timing events missing from the Events tab

Draft Role Split for Orientation

This should still be confirmed locally, but a working training draft is:

RN likely owns or co-owns

- pre-procedure readiness items
- oxygen, vitals, rhythm, pulses, pain, airway
- sedation assessment and medication administration
- postprocedure nursing assessments
- recovery progression awareness
- patient movement and handoff

Radiology Tech likely owns or co-owns

RN-Focused Epic Cupid Responsibilities

This chapter is refined primarily for the Cath Lab RN workflow. The RN should think in phases and confirm what must be charted before moving to the next phase of care.

Pre-procedure RN ownership or co-ownership

Patient in Room; patient ID verification; verifications and consents review; oxygen and pulse oximetry setup documentation; baseline vitals; ECG rhythm; pulses; pain assessment; airway assessment when sedation is planned; pre-procedure checklist completion; MD pre-sedation workflow support when applicable.

Intra-procedure RN ownership or co-ownership

Sedation assessment workflow; medication administration in MAR / meds workflow; hemodynamic and symptom surveillance; POCT / ACT awareness when assigned; real-time recognition and communication of unstable findings; confirmation that required case-tracking events are completed.

Postprocedure RN ownership or co-ownership

Site condition; pain reassessment; basic vitals; pulses; ECG rhythm when required; Aldrete score or recovery scoring when applicable; closure-related nursing awareness; patient movement / handoff completion; confirming that Case End and Sedation End are not left incomplete.

Routine Diagnostic Case Walkthrough for RN Orientation

Before patient arrival, verify the correct patient and case are open; review consent, allergies, baseline rhythm, and immediate nursing concerns; and confirm readiness of oxygen, monitoring, medications, and emergency equipment. When the patient enters the room, document or verify Patient in Room, confirm patient ID and consents, complete baseline vitals, rhythm, pulses, pain, and oxygen-related entries, complete airway assessment if sedation is planned, and confirm pre-procedure checklist readiness. At case start, ensure timeout is complete; verify In Room, Procedure Start, and Case Start workflow as applicable; document

Sedation Start if used; and chart medications in real time. During diagnostic angiography, continuously monitor symptoms, ECG, blood pressure, SpO2, and sedation response; communicate dampening, ischemic changes, hypotension, or patient symptoms immediately; and maintain awareness that angiography, catheter, and wire events are being documented appropriately. At case end, confirm debrief completion, site condition and postprocedure assessments, verify contrast, radiation, and removal events, verify Case End, Sedation End, and Out of Room when applicable, and complete patient movement and handoff.

Routine PCI Case Walkthrough for RN Orientation

Before intervention begins, confirm anticoagulation awareness and ACT plan, ensure needed medications and rescue medications are available, and maintain clear baseline hemodynamic and rhythm awareness. During PCI, chart medications in real time, maintain active surveillance for chest pain, ST change, hypotension, dysrhythmia, no-reflow concern, or access-site issues, communicate changes immediately, maintain awareness of increasing complexity or multiple device exchanges, and verify that supplies, implants, contrast, and major procedural phases are being captured. At PCI completion, reassess symptoms, vitals, pulses, and site condition, verify all wires and catheters removed documentation, confirm closure workflow, verify Case End, Sedation End, and postprocedure recovery items, and complete nursing handoff with access, sedation, and complication summary.

STEMI Workflow for RN Orientation

For emergent STEMI activation, use the emergency case-creation workflow if assigned, confirm the patient is attached to the correct live case in Epic, ensure the actual cardiologist is listed as primary and the generic provider is removed once the workflow allows, prioritize rapid but accurate nursing documentation, maintain strong awareness of hemodynamics, ischemia, arrhythmia, and patient response, and verify that major timing events are not missed, especially Procedure Start, Case Start, Case End, and Out of Room. After reperfusion or case completion, confirm prompt postprocedure site and recovery workflow and complete handoff to the next phase of care.

Advanced Coronary Documentation: FFR, iFR, IVUS, OCT

After the supply used to measure FFR, iFR, IVUS, or OCT has been inserted into the target coronary artery, the documenter expands the Diagnostic Heart Cath toolbox and selects the FFR, IVUS, & more event. The lesion entry requires vessel, segment, stenosis percent, whether FFR/iFR/IVUS/OCT was used, and measurement details. If a post-intervention reassessment is performed, the documenter opens the Coronary Intervention toolbox, selects FFR, IVUS, & more again, chooses the previously documented lesion, and enters the post-stenosis percent and post-intervention measurement details. Nursing/documentation staff should ask the cardiologist for the stenosis percentage when needed, because this lesion data can flow directly into the physician report in Study Review.

Additional Local RN Workflows

TEE/Cardioversion: the Cath Lab nurse schedules the patient from the Snapboard or depot workflow, documents In Room, opens the Procedure Log, documents Case Start, uses the

TEE/Cardioversion toolbox for the TEE portion of the case, then finishes the Procedure Log and addresses verify checks. If TEE is done with a Cath case, create a second panel for the additional procedure.

Canceling a case: Cath Lab cases should be canceled only by Cath Lab staff. Use the Events tab to document Procedure Aborted, then complete the posted/unposted log and charge-entry cancellation workflow according to local policy.

ACT / Hemochron basics: use fresh whole blood in a non-heparinized syringe, ensure the cuvette is at room temperature, add the sample and press Start within five minutes once the instrument displays Ready, avoid bubbles, and repeat results that are out of range or inconsistent with the clinical picture.

RN End-of-Case Audit

Before the patient leaves Cath Lab care, the RN should perform a final sweep to confirm timeout and debrief, In Room, Case Start, Case End, Sedation Start / Sedation End if used, Out of Room, contrast, radiation, access and sheath workflow, all wires removed, all catheters removed, site condition, closure / hemostasis workflow, postprocedure vitals, pulses, pain, recovery score as required, and patient movement / handoff.

- staff and panel support
- access procedural workflow support
- diagnostic and interventional device documentation support
- supplies and implants support
- radiation awareness and related workflow support
- technical case-end completion support

Shared / verify locally

- timeout
- case start
- procedure start and end
- closure-related entries
- contrast and removal-related completion items

Before the Patient Leaves the Room Checklist

Before transfer from Cath Lab care, verify: - timeout complete - meds charted - access events charted - sheath, catheter, and wire entries complete - supplies and implants entered - contrast documented - radiation documented - site condition documented - closure device documented if used - all catheters removed documented - all wires removed documented - postprocedure vitals and pulses documented - pain reassessed - sedation end documented if applicable - case end documented - patient movement and handoff completed

Suggested Preceptor Focus for This Tab

The orientee should be taught to think of Epic Cupid as a phase-based procedural workflow, not a collection of unrelated boxes. The most useful habit is to ask what phase the patient is in and what documentation that phase requires.

Teaching Pearls

- The left toolbox is the action list; the Event Log is the story; the Events tab is the timing control.
- Real-time charting prevents missing steps at case end.
- The fastest way to find charting gaps is to ask what phase the patient is in right now and what still has not been documented.

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12. Role-Specific Skills Checklists

Purpose

Define the practical skills each orientee must demonstrate in their primary role. These checklists are designed for use during real cases and should be completed progressively during orientation.

Core Principle

Competency should be based on observed performance, not exposure alone. Seeing a task once is not equivalent to performing it safely.

How to Use These Checklists

For each skill, the preceptor should indicate whether the orientee has: - observed the skill - performed it with guidance - performed it independently - requires remediation

A date and preceptor initials should be entered when a skill is validated.

RN Skills Checklist

A. Pre-Procedure Readiness

- verifies patient identity correctly
- confirms consent and planned procedure
- reviews allergies and key pre-procedure risks
- confirms baseline vitals and rhythm
- assesses pain and symptoms pre-procedure
- performs or verifies airway assessment when sedation is planned
- confirms IV access and medication readiness

- participates in pre-procedure checklist completion
- recognizes when readiness is incomplete and escalates before case start

B. In-Room Setup and Early Workflow

- assists with safe patient room entry and monitor hookup
- verifies oxygen and pulse oximetry setup as needed
- confirms baseline nursing assessment elements are documented
- participates in timeout appropriately
- identifies when case start documentation is required

C. Sedation and Medication Workflow

- performs sedation assessment according to workflow
- administers medications safely within role and policy
- reassesses patient after sedation or analgesia administration
- recognizes oversedation or respiratory compromise early
- maintains awareness of cumulative sedative effect during prolonged cases
- charts medication administration accurately in real time

D. Hemodynamic and ECG Monitoring

- recognizes hypotension requiring escalation
- recognizes pressure dampening and reports immediately
- identifies new ST change or unstable rhythm and escalates immediately
- tracks patient symptoms during angiography and intervention
- maintains awareness of patient responsiveness during the case

E. Anticoagulation and POCT Awareness

- understands when ACT or other POCT is due
- recognizes importance of communicating ACT timing and result
- maintains awareness of bleeding risk during and after anticoagulation
- charts POCT-related workflow accurately when assigned

F. Access and Site Monitoring

- understands basic radial versus femoral workflow differences
- monitors for access-site bleeding or hematoma
- recognizes post-closure or post-sheath-removal site concerns
- reassesses pulses and site status when required
- communicates access-related changes immediately

G. Postprocedure and Recovery Workflow

- completes or supports postprocedure nursing reassessment
- documents site condition accurately
- confirms pain reassessment and vital sign reassessment
- understands recovery phase progression in Epic Cupid
- provides or supports structured handoff to the receiving team

H. Epic Cupid RN Documentation

- enters pre-procedure nursing workflow correctly
- uses Sedation Assessment appropriately
- charts medications in the correct location
- documents postprocedure nursing elements correctly
- identifies missing charting before patient leaves Cath Lab care

I. Professional Communication and Prioritization

- uses concise call-out language during changes in patient status
- asks for clarification before guessing when safety is affected
- prioritizes patient instability over lower-level tasks
- remains organized during rapid workflow change

Radiology Tech Skills Checklist

A. Room and Equipment Readiness

- prepares the room before patient arrival
- confirms imaging system readiness
- organizes table setup and procedural supplies appropriately
- verifies shielding and detector readiness
- locates emergency and bailout equipment rapidly

B. Sterile Workflow and Field Protection

- maintains sterile field awareness
- identifies and reports sterile breaks immediately
- supports organized supply handling during routine and complex cases
- maintains field integrity during rapid equipment exchange

C. Procedure Flow Support

- identifies major phases of a routine cath or PCI case
- anticipates common wires, catheters, balloons, and stents in routine workflow
- supports efficient procedural sequencing without losing organization
- recognizes when the case is escalating in complexity

D. Radiation and Imaging Awareness

- maintains awareness of radiation exposure during the case
- supports detector and imaging optimization
- recognizes when prolonged fluoroscopy or imaging burden is occurring
- participates in workflow that reduces unnecessary exposure

E. Access and Device Workflow Support

- identifies basic access equipment and sheath workflow
- supports wire, catheter, and device readiness

- recognizes abnormal resistance as a safety concern
- remains attentive during exchanges and complex device use

F. Supplies and Implants Documentation Support

- enters or supports supply documentation accurately as assigned locally
- distinguishes used versus wasted supply workflow
- identifies when implant documentation is required
- audits supply completion before case closure when assigned

H. Epic Cupid "Before Patient Leaves" Checklist

- identifies major Procedure Log sections
- uses the left-side toolbox in workflow order
- supports access, procedure, and postprocedure entries as assigned locally
- understands the Events tab role in case timing
- identifies missing technical charting elements before patient transfer

H. Complex Case Support

- anticipates increased equipment needs in calcified, bifurcation, left main, thrombotic, or prolonged cases
- maintains organization during repeated device exchange
- protects sterile flow under pressure
- responds effectively to urgent procedural requests without losing accuracy

I. Communication and Team Function

- calls out technical or workflow concerns clearly
- communicates equipment readiness accurately
- reports resistance, equipment concern, or sterile issue immediately
- functions within role without delay during unstable events

Shared Competencies for All Cath Lab Staff

- recognizes the current phase of the case
- understands when a case has become nonroutine
- escalates critical changes immediately
- maintains calm and role clarity during complications
- supports safe patient transfer and complete documentation

Skills Validation Table

Skill	Observed	Performed with Guidance	Performed Independently	Remediation Needed	Date	Preceptor Initials

13. Quick Reference Appendix

Purpose

Provide fast-access reference tools for high-yield daily workflow, common call-outs, and end-of-case safety checks. This section should eventually be converted into one-page laminated guides for room use.

Core Principle

Quick references should support rapid action and consistency. They should not replace training, but they should reduce omission and hesitation during live cases.

Appendix Sections to Include

A. Pre-Case Readiness Quick Check

Before timeout, confirm: - patient identity verified - consent verified - procedure confirmed - allergies reviewed - baseline vitals documented - rhythm documented - airway assessed if sedation planned - access plan known - emergency equipment available - monitoring active - Epic Cupid baseline documentation started

B. Standard Case Phase Reference

1. Pre-procedure
2. In room
3. Timeout
4. Case start
5. Procedure start
6. Intra-procedure
7. Postprocedure
8. Case end
9. Recovery / transfer

C. Critical Call-Out Reference

Immediate call-outs include: - new ST changes - pressure dampening - hypotension - oxygen desaturation - site bleeding - ACT due - contrast rising - radiation rising - resistance advancing device - possible perforation or dissection

D. Angiography Sequence Quick Reference

- confirm safe engagement
- check waveform
- test injection if needed
- reassess ECG and patient response

- complete cine run
- reassess flow, ECG, pressure, and symptoms

E. PCI Phase Reference

- planning
- access
- engagement
- wiring
- lesion preparation
- imaging or physiology
- stent delivery
- optimization
- final assessment
- closure

F. Access and Closure Quick Check

- site confirmed
- sheath size known
- wire/catheter events documented
- closure plan known
- site reassessed after closure or sheath removal
- hemostasis documented if applicable

G. Cath Lab Medications Quick Sheet

Sedation / reversal

Midazolam: 0.5-1 mg IV, repeat as needed; watch for respiratory depression; flumazenil reversal.

Fentanyl: 25-100 mcg IV, repeat as needed; watch for respiratory depression and delayed oral P2Y12 effect; naloxone reversal.

No-reflow / spasm / vasodilators

Nitroglycerin: infusion 10 mcg/min, increase by 10 mcg/min every 5 min; avoid hypotension, RV infarct, PDE-5 exposure, and HOCM.

Nicardipine IC: 100-300 mcg.

Nitroprusside IC: 100-300 mcg.

Verapamil: radial 2-3 mg; IC 1 mg over 2 min.

Adenosine: RCA 50-100 mcg IC; left main 100-200 mcg IC; monitor for AV block and arrhythmia.

Anticoagulation / ACT

ACT every 20-30 min.

Goal ACT without GP IIb/IIIa or cangrelor: Hemochron 300-350 sec.

Goal ACT with GP IIb/IIIa or cangrelor: 200-250 sec.

Protamine: 1-1.5 mg per 100 units heparin; max 50 mg; give slowly.
Bivalirudin: 0.75 mg/kg bolus, then 1.75 mg/kg/h infusion; no reversal.

Antiplatelets

Aspirin 325 mg load -> 81-100 mg daily.
Clopidogrel 600 mg load -> 75 mg daily.
Ticagrelor 180 mg load -> 90 mg twice daily.
Prasugrel 60 mg load -> 10 mg daily, 5 mg if under 60 kg or over 75 years.
Cangrelor 30 mcg/kg bolus -> 4 mcg/kg/min infusion.

H. Epic Cupid "Before Patient Leaves" Checklist

- In Room documented - timeout complete - meds charted - access documented - supplies entered - contrast documented - radiation documented - site condition documented - all catheters removed documented - all wires removed documented - Case Start completed - Case End completed - Sedation End completed if applicable - Out of Room documented - patient movement and handoff complete
- meds charted
- access documented
- supplies entered
- contrast documented
- radiation documented
- site condition documented
- all catheters removed documented
- all wires removed documented
- case end completed
- sedation end completed if applicable
- patient movement and handoff complete

I. Events Tab Timing Audit

- in facility - in pre-procedure - pre-procedure complete - in room - procedure start - case start if required - case end - procedure finish - sedation end if applicable - out of room - recovery progression documented
- in pre-procedure
- in room
- procedure start
- case start if required
- case end
- procedure finish
- sedation end if applicable
- out of room
- recovery progression documented

J. Postprocedure Handoff Quick Reference

Include at minimum: - procedure performed - access site and closure method - sedation received - complications or concerns - monitoring priorities - postprocedure restrictions - pending orders or follow-up tasks

K. STEMI / Emergency Case Quick Reference

Use the STEMI button from the toolbar for emergent case creation; enter MRN and select patient; create case to place it on the status board; update the physician to the actual cardiologist and mark Primary; remove the generic provider; update or add procedures as needed; continue standard Procedure Log documentation once the case is live.

L. Complication Recognition Quick Reference

Include one-line triggers for: - acute vessel closure - dissection - no-reflow - perforation - air embolism - vasovagal event - access-site bleeding - hemodynamic collapse

Recommendation for Final Design

Each of the above should eventually become a separate one-page sheet with bold headings, minimal text, and highly visible action points.

Recommended Priority for Quick Reference Buildout

Create these first because they are most likely to be used during active workflow: 1. Epic Cupid “Before Patient Leaves” checklist 2. Critical call-out reference 3. Pre-case readiness quick check 4. PCI phase reference 5. Access and closure quick check 6. Postprocedure handoff quick reference

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14. Competency Validation and Signoff

Purpose

Provide a formal record of orientation progress and validate that the orientee can participate in Cath Lab workflow safely and consistently.

Core Principle

Competency validation should reflect real performance over time, not a single isolated observation. Final signoff should indicate readiness for routine independent participation within role.

Required Global Competencies

Before final orientation signoff, the orientee should demonstrate: - room readiness awareness - sterile awareness - monitoring vigilance - clear escalation of critical changes - accurate Epic

Cupid charting within role - safe participation in routine diagnostic cases - safe participation in routine PCI workflow - awareness of access and closure priorities - organized response to complications within role - safe postprocedure transition and handoff

Competency Categories for Final Signoff

1. Room Setup and Pre-Procedure Readiness

- understands room flow
- verifies readiness before case start
- identifies missing safety elements before procedure begins

2. Monitoring and Intra-Procedure Safety

- identifies abnormal patient, ECG, and pressure findings
- escalates immediately when indicated
- remains attentive throughout the case

3. Procedure Support Workflow

- follows the sequence of routine angiography and PCI
- anticipates the next step in routine cases
- remains organized during device exchange or workflow change

4. Epic Cupid Documentation

- charts in correct workflow sections
- uses timing, access, postprocedure, and case-end sections appropriately
- identifies omissions before transfer

5. Postprocedure and Handoff

- supports safe transfer
- completes or verifies required postprocedure workflow
- communicates access, sedation, and complication-related concerns clearly

Final Validation Scale

Use the following scale for each category: - **N/O** = Not Observed - **D** = Developing - **C** = Competent with Guidance Available - **I** = Independent for Routine Workflow - **R** = Remediation Needed

Final Competency Summary Table

Category	N/O	D	C	I	R	Date	Preceptor Initials
Room Setup and Pre-Proced							

Category	N/O	D	C	I	R	Date	Preceptor Initials
ure Readiness							
Monitoring and Intra-Procedure Safety							
Procedure Support Workflow							
Epic Cupid Documentation							
Postprocedure and Handoff							

Preceptor Signoff Template

Skill/Topic: _____
Observed Date: _____
Performed With Guidance: _____
Performed Independently: _____
Preceptor Initials: _____
Comments: _____
_____

Weekly Progress Review Template

Orientee Name: _____
Role: RN / Radiology Tech
Week of Orientation: _____
Primary Preceptor: _____

Strengths this week: _____
_____

Safety concerns noted: _____
_____

Charting gaps noted: _____
_____

Workflow gaps noted: _____
_____

Focus for next week: _____
_____

Preceptor initials/date: _____

Orientation Progress Record

Week	Focus	Status	Preceptor Initials
1	Observation / room flow / monitoring basics		
2	Pre-procedure prep / basic participation		
3	Routine diagnostic case support		
4	Basic charting / medication workflow		
5	Routine PCI participation		
6	Access and closure workflow		
7	Complication recognition / escalation		
8	Independent routine workflow validation		

Final Orientation Recommendation

At the end of orientation, the preceptor and educator should select one: - Ready for independent routine assignment - Ready for independent routine assignment with focused follow-up - Extend orientation - Not yet ready for independent assignment

Final comments: _____
_____
_____

Primary preceptor: _____
Educator / supervisor: _____
Date: _____

Final Editor Notes for Department Customization

Before final release, this binder should be reviewed for: - local medication standards and dose references - local access and closure practice patterns - exact RN versus radiology tech documentation ownership - local Epic Cupid permissions and workflow expectations - emergency contact pathways and escalation chain - department-specific policy language

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Next Revision Priorities

1. Add department-specific medication, supply, and closure pages
2. Add annotated Epic Cupid screenshots inside Tab 11
3. Convert appendix tools into laminated one-page references
4. Insert local escalation chain, phone numbers, and policy contacts
5. Apply final layout and visual design for print and digital use

Local Cath Lab Medication and Mechanical Support Addendum

Use this addendum as a local quick-reference supplement for Ochsner Lafayette General Cath Lab orientation. These notes should be verified against the active formulary, pump library, physician order, and current departmental policy before independent use.

Local Infusion and Sedation Reference

Medication / Mix	Local quick note	RN monitoring / caution
Nitroglycerin (Tridil)	Low dose 5-50 mcg/min; increase by 5 mcg q5 min. High dose 50-100 mcg/min. Premixed in glass. Example rates: 5 mcg/min = 3 mL/hr, 10 = 6 mL/hr, 15 = 9 mL/hr.	Watch for hypotension, headache, and worsening RV infarct physiology.
Nicardipine	Premixed 20 mg/200 mL D5W. Start 5 mg/hr. Titrate q10-15 min by 1-2.5 mg/hr to 2.5-15 mg/hr.	Watch BP trend and reflex tachycardia; reassess after each titration.
Nitroprusside (Nipride)	50 mg/250 mL D5W. 0.5-10 mcg/kg/min. Adjust by 0.5 mcg/kg/min q3-5 min. Protect from light. Do not run with other drugs in the same line.	Watch for hypotension and line compatibility issues.
Phenylephrine (Neosynephrine)	10 mg/250 mL D5W. Rapid support 100-180 mcg/min. Maintenance 40-60 mcg/min. Weight-based option starts 0.5 mcg/kg/min.	Watch for bradycardia, hypertension, and extravasation.
Norepinephrine (Levophed)	Single strength 4 mg/250 mL D5W = 16 mcg/mL. Double strength 8 mg/250 mL D5W = 32 mcg/mL. Start 1 mcg/min and titrate q5 min.	Watch MAP, rhythm, peripheral perfusion, and line integrity.
Vasopressin	100 units/100 mL NS. Start 0.04 units/min. Titrate by 0.01 units/min q10-15 min. Max 0.07 units/min.	Watch MAP and ischemic complications.
Dopamine	400 mg/250 mL = 1600 mcg/mL. Low dose 0.5-3 mcg/kg/min, medium 5-10,	Watch rhythm, tachycardia, and BP response.

	high 10-20. Increase by 2 mcg/kg/min q10-30 min.	
Dobutamine	500 mg/250 mL D5W = 2000 mcg/mL. Start 2-3 mcg/kg/min; increase by 2-3 mcg/kg/min q10-30 min. Challenge sheet starts 5 mcg/kg/min x3 min, then 10 mcg/kg/min x3 min, then per physician order.	Watch tachyarrhythmia, hypotension, chest pain, and symptom response.
Lidocaine infusion	Bolus 1 mg/kg (50-100 mg), infusion 1-4 mg/min. Single strength: 1 g/250 mL or 2 g/500 mL D5W. Double strength: 2 g/250 mL or 4 g/500 mL D5W.	Watch mental status, bradycardia, QRS change, and toxicity.
Dexmedetomidine (Precedex)	400 mcg total volume 100 mL NS. Loading dose 1 mcg/kg over 10 min. Infusion 0.2-0.7 mcg/kg/hr.	Watch for bradycardia and hypotension.
Sedation reversal	Flumazenil 0.2 mg IVP to start. Naloxone 0.4-2 mg IVP q2-3 min.	Reassess airway, RR, mental status, and recurrence of sedation.
Insulin drip	100 units regular insulin in 100 mL NS. 1 unit = 1 mL. 10 units/hr = 10 mL/hr.	Follow glucose protocol and potassium monitoring.

Local Device / Vessel Cocktails

Cocktail	Local note
Diamondback - Peripheral	Mix in 1000 mL 0.9% NS: verapamil 5 mg, nitroglycerin 5 mg (50 mL), ViperSlide 20 mL.
Diamondback - Coronary	Mix in 1000 mL 0.9% NS: nitroglycerin 2.5 mg (25 mL), ViperSlide 20 mL. No verapamil.
Rotablator cocktail	Mix in 1000 mL NS: nitroglycerin 5 mg (50 mL), Rotaglide 20 cc, heparin 5000 units, verapamil 10 mg = 4 mL, or per physician preference.

Always confirm local physician preference, active order set, and device-specific policy before preparation.

Mechanical Circulatory Support Quick Reference

Device	Core RN focus	Call immediately for
IABP	Confirm timing and augmentation, monitor console alarms, perform frequent bilateral pulse and limb checks, keep site visible, maintain straight-leg/activity precautions per protocol.	Loss of pulses, limb ischemia signs, poor augmentation, major bleeding/hematoma, blood in tubing, neurological change.
Impella	Monitor performance level, flow, purge pressure/flow, alarms, large-bore access site, bilateral pulses, urine color, hemolysis/AKI concerns, and anticoagulation awareness.	Suction alarms, low flow, pink/red urine, worsening hypotension, purge problem, new ventricular arrhythmia, loss of pulses, suspected malposition.

Addendum Use Notes

This addendum is intended as a local quick-reference supplement and should not replace active physician orders, the pump library, Epic MAR instructions, or current Ochsner / OLG policy.

Where this addendum differs from the active order, formulary, or pump library, the active order and pump library should be followed.

Consider printing these pages as a local appendix or laminating selected tables for controlled room use after educator review.